Synopsis: Are India's Monsoon Contingency Plans Still Right?

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V Semester Experiential Learning · SDG 13: Climate Action · R V College of Engineering · Team of 3–4 · Software only

This project checks whether India's District Agriculture Contingency Plans, written mostly in 2011–2012, still fit today's rainfall and crops. LLM agents split 105 district plans from four states into testable claims. They check each claim with code tools against IMD daily rainfall (1981–2025) and district crop statistics (1997–2023).

A verifier agent re-checks every cited number, and the whole system is compared with a rule-based checker and a single LLM prompt. The output is a "plan freshness" score for every district, evidence-cited draft updates for experts, and a map dashboard.

1. Background and motivation

Almost every Indian district has a District Agriculture Contingency Plan that says what farmers should sow and do when the monsoon goes wrong. ICAR-CRIDA and agricultural universities have prepared about 650 such plans. Each covers the same situations: monsoon delays of 2–8 weeks, early, mid and late-season drought, floods and heat waves.

The plans rest on old data. Most of the 105 plans used here are dated 2011–2012, and the five we opened use crop figures from 2004–2011. Rainfall and cropping may have shifted since then, but we found no study that checks the plans against newer data at scale.

The need is current. On 23 June 2026, with about 315 districts flagged for a possible weak monsoon, the Union Agriculture Minister asked states to review and update these plans for local conditions (PIB).

2. Problem statement

Contingency plans guide what farmers sow in a bad monsoon year, but they were written for the rainfall and crops of the 2000s. There is no systematic, data-driven way to tell which parts of which plans are now out of date.

The problem: turn each plan into claims that can be tested, check every claim against recent rainfall and crop data without inventing evidence, and show experts which districts and sections need revision first.

Research question: can tool-using LLM agents, which turn plan text into testable claims and check them against rainfall and crop records, find outdated plan assumptions more accurately than a rule-based checker or a single LLM prompt?

3. Objectives

1. Collect 105 district plans from Karnataka (29), Maharashtra (34), Telangana (9) and Rajasthan (33), and link each to its Census 2011 district.
2. Build evidence tools in code: district rainfall, sowing-rain onset, dry spells, crop-area trends and drought-year crop shifts, comparing 1981–2010 with 2011 onward.
3. Build a multi-agent system that splits each plan into testable claims, checks each claim with the tools, labels it supported, contradicted or not enough evidence, and verifies every cited number.
4. Build an evaluation set: about 300 hand-labelled claims from 30 plans, plus planted errors with known answers.
5. Compare the agents with a rule-based checker and a single LLM prompt, and measure what the verifier adds.
6. Turn the claim labels into a plan freshness score per district with draft updates, show them on a map dashboard, and release the claim dataset and code.

4. Novelty and research gap

Existing tools use the plans as written; none tests whether their assumptions still hold. This project adds that test, claim by claim, against long-term data. Multi-agent systems and claim checking are not new; using them to find outdated plan assumptions is the contribution.

What exists	What it lacks	What this project adds
District contingency plans (ICAR-CRIDA, about 650)	Static PDFs, mostly from 2011–2013; no routine check against new data	Plan contents as testable claims, each checked against data from 2011 onward
CRIDA's online contingency tool	Shows a plan beside current rainfall; does not test the plan	Tests whether each plan's assumptions still hold
SukhaRakshak AI (IWMI, 2025), a chatbot that answers from the plans	Repeats the plans' advice as written	Flags where that advice may be out of date, with the evidence

What exists	What it lacks	What this project adds
Claim-checking research: FEVER and Climate-FEVER (claims against text), TabFact (claims against tables), FActScore (splitting text into small facts)	Checks claims against text or a single table, not long-term records	Checks plan claims against 45 years of rainfall and 26 years of crop data to measure drift
Rainfall trend studies on IMD data	Describe climate change, but are not tied to farm advice	Links rainfall shifts to specific plan triggers and actions

5. Datasets

All data is Indian, free and already checked; no satellite data is used. The full download is about 1–1.5 GB, so a laptop is enough.

Dataset	Role	Coverage	What it holds (checked)	Access
District Agriculture Contingency Plans (ICAR-CRIDA)	What is being checked	105 districts in 4 states; 12 Andhra Pradesh plans optional	One PDF per district, same template; sample links opened; all matched to Census 2011 districts	Free PDFs
IMD daily gridded rainfall, 0.25°	Rainfall check	1981–2025 used (1901 onward exists)	Daily rain in mm on a 135 × 129 grid, about 24 MB a year; every plan district has at least 3 grid points (median 11)	Free download
District crop area, production and yield (Ministry of Agriculture, via the India Data Portal)	Crop and uptake checks	1997–98 to 2022–23	District × season × crop for all of India, one CSV; units given on every row	Free, no login

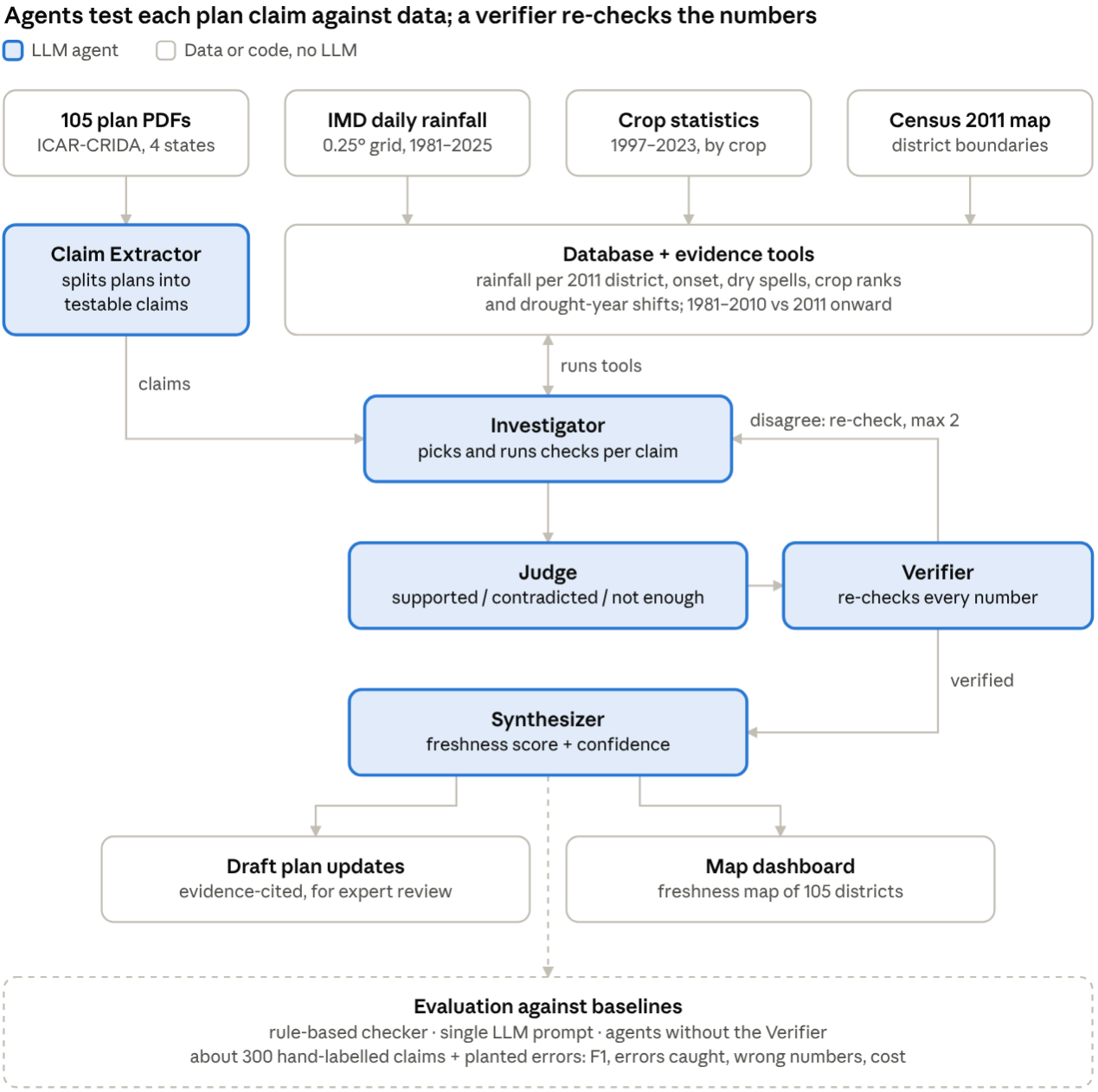
Dataset	Role	Coverage	What it holds (checked)	Access
Census 2011 district boundaries (DataMeet)	Joins everything by district	All 2011 districts	Same districts as the plans; Palghar is inside Thane, Telangana under Andhra Pradesh	Free (CC BY 2.5 India)

Districts formed after 2011 are added back into their 2011 parent, so every source uses the same district list.

No ground truth for "outdated" exists, so the project builds its own evaluation set. It has about 300 claims from 30 plans, hand-labelled by the team, plus planted errors whose answers are known, such as a crop the district does not grow or a sowing date moved by 6 weeks.

6. Methodology

Code tools compute all the evidence; LLM agents decide which evidence each claim needs, judge it, and check each other's work. Steps 1–10 are the core project; stretch goals come only after them.



method · 5 LLM agents over code tools, compared with 3 baselines

The agents never compute numbers themselves: the Investigator calls code tools, and the Verifier sends any claim it disagrees with back for another check.

1. **Week-1 gate.** Download the 105 plans and confirm at least 100 are readable. Run the agents on 5 plans to measure accuracy, time and LLM cost before building further.
2. **Database.** Load plan text, district rainfall (IMD grid points averaged over each 2011 district, weighted by area) and crop statistics into PostgreSQL with PostGIS.
3. **Evidence tools (code, no LLM).** Functions for June–September totals, sowing-rain onset, dry spells, heavy-rain days, crop-area ranks and crop shifts in deficient years (rain at least 20% below normal, IMD's definition). Each compares 1981–2010 with 2011 onward, using non-parametric tests corrected for testing many districts.

4. **Claim Extractor agent.** Splits each plan into small testable claims with page references: normal crops, sowing windows, trigger thresholds (for example, a 15–20-day dry spell after sowing) and recommended actions.
5. **Investigator agent.** For each claim, chooses which tools to run, runs them, and tries again when data is missing, for example by using the 2011 parent of a split district.
6. **Judge agent.** Labels each claim supported, contradicted or not enough evidence, citing the tool outputs.
7. **Verifier agent.** Re-runs the cited checks on its own and compares every number. Disagreements go back to the Investigator, at most twice.
8. **Synthesizer agent.** Combines the claim labels into a 0–100 freshness score per district with a confidence level, and drafts updates for contradicted claims, citing their evidence.
9. **Evaluation.** Compare four systems on the evaluation set: a rule-based checker, one LLM prompt given the plan and a data summary, the agents without the Verifier, and the full system. Measure label F1, planted errors caught, wrong numbers per 100 claims, citation accuracy and cost per plan.
10. **Dashboard.** A map of freshness scores, with each district's claims, evidence and draft updates one click away.

Stretch goals: extend the check to all 650 plans, test the heat-wave sections with IMD temperature data, and have an agronomist review about 20 district results.

7. Tech stack

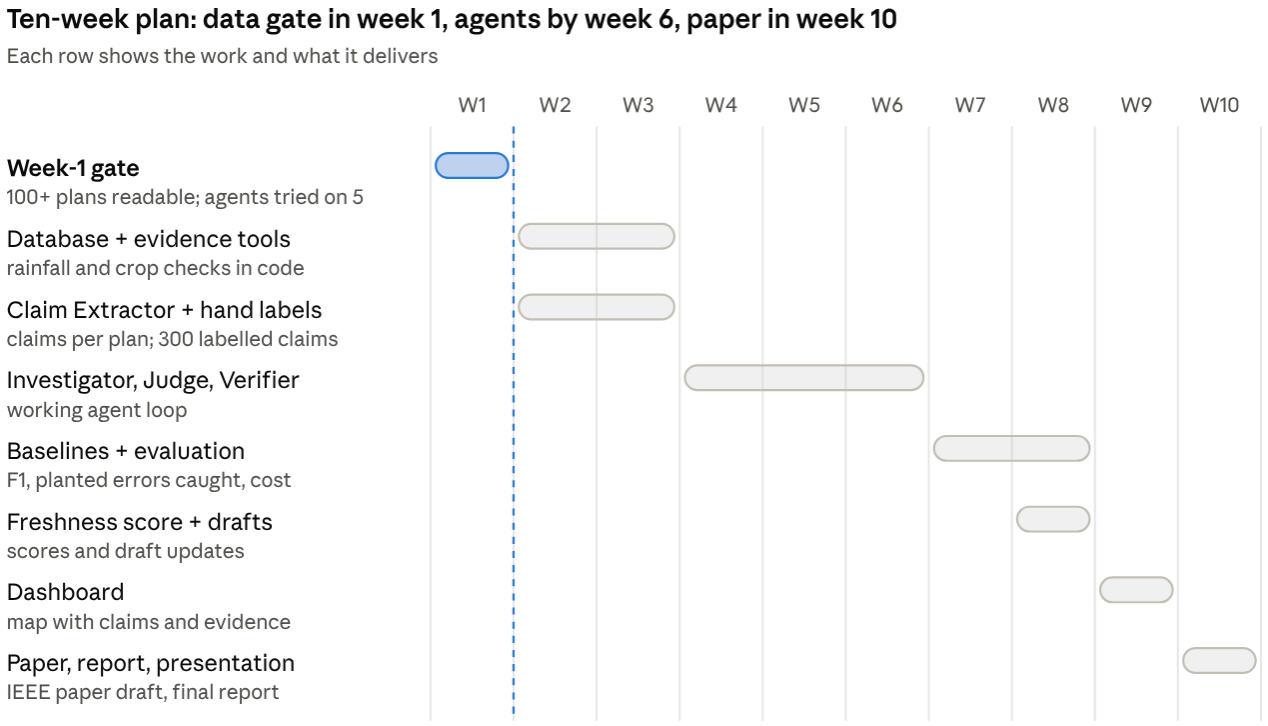
Every tool is free and runs on a laptop or Google Colab. The only possible cost is an LLM API, if one is used instead of a local model.

Layer	Tools
Language	Python 3
PDF to text	PyMuPDF; Tesseract OCR for scanned pages
LLM agents	An LLM with tool calling (open-weight via Ollama, or an API); LangGraph or plain Python for the agent loop; Pydantic for JSON outputs
Database	PostgreSQL with PostGIS
Rainfall and maps	xarray, netCDF4, geopandas, xagg (area-weighted grid-to-district averages)

Layer	Tools
Statistics and evaluation	pandas, SciPy, statsmodels, pymannkendall, scikit-learn
Dashboard	Streamlit, Folium
Collaboration	GitHub, Google Colab

8. Timeline

The work fits in 10 weeks, and nothing is built until the week-1 gate passes.



work plan · 10 weeks, 8 work packages

Evidence tools and claim extraction run in parallel in weeks 2–3, so the team can split the work.

9. Expected outcomes

The main outcome is an IEEE conference paper, backed by a working tool and an open dataset.

- **Paper:** whether tool-using agents find outdated plan claims better than a rule-based checker or a single LLM prompt, how far plans in four states have drifted since 2011, and which districts need revision first. Every result is reportable: if the agents do not

beat the rules, or most plans still fit, that is still evidence.

- **Product:** a freshness map and evidence-cited draft updates that CRIDA, agricultural universities and state agriculture departments can use before each monsoon.
- **Open release:** the claim dataset (about 300 hand-labelled claims plus planted errors), the district matching table and the code on GitHub.
- **SDG 13:** target 13.1 (resilience to climate hazards) and target 13.2 (climate measures in planning).

10. Risks and mitigation

Verdict: doable. All data sources are checked. The two main risks, reading the PDFs and making the agents reliable, are tested on 5 plans in week 1. The rule-based checker means the project finishes either way.

Risk	Likelihood	Mitigation
No existing ground truth for "outdated"	Certain	Build it: about 300 hand-labelled claims and planted errors with known answers; agronomist review as a stretch goal
Agents use tools unreliably, or cost too much	Medium	Accuracy and cost tested on 5 plans in week 1; the rule-based checker stays in the core
LLM misreads messy or scanned plan tables	Medium–high	OCR, fixed claim types, page references; extraction checked against the hand-labelled claims
LLM agents invent numbers or advice	Medium	Numbers come only from code tools; the Verifier re-runs every cited check; drafts are for expert review only
Fewer readable plans than expected	Low	105 links already verified; week-1 gate of 100 readable plans; Andhra Pradesh plans as reserve
District boundaries changed after 2011	Certain	Census 2011 map; new districts merged into their 2011 parent; matching table already built
Crop-response results read as cause and effect	High	Reported as association; before-plan versus after-plan drought years compared

Risk	Likelihood	Mitigation
Gaps in crop data	Medium	Focus on major crops; coverage reported per district
Rain-gauge network behind the IMD grid changes over time	Medium	Stated as a limitation; the same dataset is used for both periods
Scope too large for the semester	Medium	Core steps first; stretch goals only if time is left

11. References

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